

From (iii), we have: $c = 20 - (e + f + g)$
 $= 20 - (2 + 9 + 3) = 20 - 14 = 6.$

From (ii), we have: $b = 30 - (d + f + g)$
 $= 30 - (7 + 9 + 3) = 30 - 19 = 11.$

From (i), we have: $a = 40 - (d + e + g)$
 $= 40 - (7 + 2 + 3) = 40 - 12 = 28.$

$\therefore$ Number of aliens who had one or more of the three unusual features

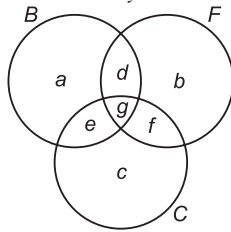
$$= a + b + c + d + e + f + g$$

$$= (28 + 11 + 6 + 7 + 2 + 9 + 3) = 66.$$

Hence, number of aliens without any of the unusual features $= (100 - 66) = 34.$

Questions 462 to 466

Let the circles B, F and C represent the children who play badminton, football and cricket respectively. Then, $c = 5, b = 8, a = 7, d = 3, f = 4, e = 4, g = 2.$



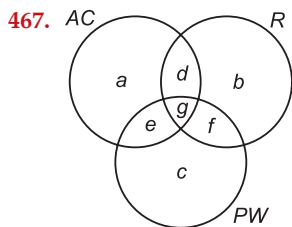
462. Required number $= f + g = 4 + 2 = 6.$

463. Required number $= a + d + e + g = 7 + 3 + 4 + 2 = 16.$

464. Required number $= b + d + f + g = 8 + 3 + 4 + 2 = 17.$

465. Required number $= e + g = 4 + 2 = 6.$

466. Required number $= a + b + c + d + e + f + g$
 $= 7 + 8 + 5 + 3 + 4 + 4 + 2 = 33.$



$$a + d + e + g = 15 \quad \dots(i)$$

$$e = 2 \quad \dots(ii)$$

$$b + d + f + g = 12 \quad \dots(iii)$$

$$d = 6 \quad \dots(iv)$$

$$c + e + f + g = 11 \quad \dots(v)$$

$$f + g = 4 \quad \dots(vi)$$

$$g = 3 \quad \dots(vii)$$

From (i), (ii), (iv) and (vii), we have: $a = 15 - (d + e + g)$
 $= 15 - (6 + 2 + 3)$
 $= 15 - 11 = 4.$

From (vi), we have: $f = 4 - g = 4 - 3 = 1.$

From (iii), we have: $b = 12 - (d + f + g)$
 $= 12 - (6 + 1 + 3) = 2.$

From (v), we have: $c = 11 - (e + f + g)$
 $= 11 - (2 + 1 + 3) = 5.$

$\therefore$ Number of cars which had at least one option
 $= a + b + c + d + e + f + g$
 $= 4 + 2 + 5 + 6 + 2 + 1 + 3 = 23.$

So, number of cars that had none of the options
 $= 25 - 23 = 2.$

468. $a + d + e + g = 23 \quad \dots(i)$

$$b + d + f + g = 15 \quad \dots(ii)$$

$$c + e + f + g = 18 \quad \dots(iii)$$

$$d = 3 \quad \dots(iv)$$

$$f = 6 \quad \dots(v)$$

$$e = 6 \quad \dots(vi)$$

$$a = 9 \quad \dots(vii)$$

From (i), we have: $g = 23 - (a + d + e)$
 $= 23 - (9 + 3 + 6) = 5.$

$\therefore$ Number of students who speak all the three languages $= g = 5.$

469. $a + d + e + g = 200 \quad \dots(i)$

$$b + d + f + g = 150 \quad \dots(ii)$$

$$c + e + f + g = 150 \quad \dots(iii)$$

$$d + g = 80 \quad \dots(iv)$$

$$f + g = 60 \quad \dots(v)$$

$$e + g = 70 \quad \dots(vi)$$

$$a = 80 \quad \dots(vii)$$

From (i), (iv) and (vii), we get:

$$a + e = 200 - (d + g) = 200 - 80 = 120.$$

$$\Rightarrow e = 120 - a = 120 - 80 = 40.$$

From (vi), we have: $g = 70 - e = 70 - 40 = 30.$

From (v), we have: $f = 60 - g = 60 - 30 = 30.$

From (iv), we have: $d = 80 - g = 80 - 30 = 50.$

From (ii), we have: $b = 150 - (d + f + g) = 150 - (50 + 30 + 30) = 40.$

From (iii), we have: $c = 150 - (e + f + g)$
 $= 150 - (40 + 30 + 30) = 50.$

Number of Dr Hormis' patients $= b = 40.$

Number of Dr Gerard's patients $= c = 50.$

Number of Dr Paul's patients $= d + e + f + g$
 $= 50 + 40 + 30 + 30 = 150.$

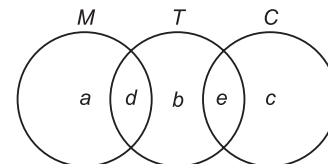
470. $a + b + c + d + e = 18 \quad \dots(i)$

$$a + d = 10 \quad \dots(ii)$$

$$b + d + e = 8 \quad \dots(iii)$$

$$c + e = 6 \quad \dots(iv)$$

$$d = 5 \quad \dots(v)$$



Adding (ii) and (iv) and subtracting from (i), we get:

$$b = 18 - (10 + 6) = 18 - 16 = 2.$$

From (iii), we have:

$$e = 8 - (b + d) = 8 - (2 + 5) = 8 - 7 = 1.$$

$\therefore$ Number of members who take tea as well as coffee
 $= e = 1.$

Questions 471 to 475

We have: $a = 650, b = 550,$

$c = 450, g = 100, d + g = 200,$

$f + g = 400, e + g = 300.$

So,

$e = 300 - g = 300 - 100 = 200; f$

$= 400 - g = 400 - 100 = 300;$

$d = 200 - g = 200 - 100 = 100.$

